

ACHINTYA RAI

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EDUCATION

University of California, San Diego
Bachelor of Science in Computer Engineering

Sept. 2024 – June 2028
GPA: 3.9/4.0

WORK EXPERIENCE

Adobe

Software Engineering Intern

San Jose, CA

June 2026 – Sept. 2026

- Profiled **60.6M+** payment events in Databricks, flagging **86,387** duplicate records and a **47%** timestamp null rate to establish the source of truth for downstream real-time monitoring.
- Designed a time-series anomaly detector handling multi-layer seasonality across **30+** countries.
- Deployed MLOps and SQL pipelines to retrain and serve machine learning models in production.

VALT Health

Founding Software Engineer

Remote

Aug. 2025 – Feb. 2026

- Engineered a Flask backend ingesting **100k+** datapoints, unifying **20+** sources into one schema.
- Implemented async batch pipelines in GCP, boosting throughput **35%** and cutting writes **60%**.
- Automated local environment setup and data seeding via Bash, cutting testing time by **80%**.

Decklar (formerly Roambee)

Software Engineering Intern

Sunnyvale, CA

June 2024 – Aug. 2024

- Developed real-time monitoring modules for **Amazon**, **Coca-Cola**, and **Samsung**, optimizing Python and NumPy data pipelines for a **15%** throughput boost on **1M+** weekly events.
- Implemented advanced data structures and algorithms, cutting analytics latency by **20%**.
- Refined Panels dashboards via iterative user testing, lifting workflow efficiency by **15%**.

RESEARCH

Graph Theory & Neural Architecture

Undergraduate Research Fellow (NSF REU)

San Diego, CA

March 2026 – Present

- Engineered a Python pipeline to analyze **LLM** embedding manifolds (Llama-2, DeepSeek-V2), building weight-matrix extraction utilities that compressed **500GB** checkpoints into **13GB** manifolds.
- Optimized placement and preprocessing scripts for a **30x** runtime cut on **8GB** RAM systems.
- Applied Rent's Rule and Louvain clustering to raise Inter-Cluster density to **13.25%** in Llama-2.

Applications of Multi-Agent Systems

Undergraduate Research Fellow (UC Scholars)

San Diego, CA

June 2025 – Aug. 2025

Multi-agent AI Chess Tutor | *Gemini API, MCP, Django, React, SQLite*

- Architected multi-agent LLM workflows that cut hallucinations **80%** while running **83%** faster.
- Engineered custom **MCP** servers and tools, standardizing tool access across an agent framework.
- Built the AI chess analysis tool with **Django**, **React**, and **SQLite**, validated by **15+** coaches.

PROJECTS

NanoGPT | *Python, PyTorch, Transformers, Neural Networks* 

Mar. 2026

- Built a transformer from scratch in **PyTorch** with multi-head attention and backpropagation.
- Trained a **100k+**-parameter character-level language model at **100%** GPU utilization.

Watchdog | *Python, Gemini API, Browser-Use, Asyncio, SQLite* 

April 2026

- Engineered an agentic browser safeguard with **Gemini Vision** to autonomously monitor tabs.
- Built a concurrent **Asyncio** CLI for real-time URL blacklisting and **Notion API** task syncing.

SquaredUpChess | *React, TypeScript, Django, WebSockets, AWS, Docker* 

Aug. 2025 – Dec. 2025

- Engineered a full-stack **AWS** platform with a **React** UI delivering low-latency, real-time updates.
- Established a **CI/CD** pipeline with automated testing, sustaining **99.9%** deployment reliability.

TECHNICAL SKILLS

Languages: Python, Java, C++, C, TypeScript, JavaScript, SQL (Postgres/SQLite), Bash, HTML/CSS

Frameworks: React, Django, FastAPI, Flask, Node.js, NumPy, Asyncio, WebSockets, PyTest, JUnit

ML & Cloud: PyTorch, TensorFlow, LLMs, MLOps, AWS, GCP, Databricks, Gemini/OpenAI API, MCP, Docker